

1. priors $p(-) = \frac{3}{5}$ $p(+)=\frac{2}{5}$ $|V|=20$

$P_{w|c}$ negative $n_- = 14$
positive $n_+ = 9$

$$P(w_i/c) = \frac{\text{count}(w_i, c) + 1}{(\sum_{w \in V} \text{count}(w, c) + |V|)}$$

$$P(\text{"predictable"}|-) = \frac{1+1}{14+20}$$

$$P(\text{"no"}|-) = \frac{1+1}{14+20}$$

$$P(\text{"fun"}|-) = \frac{0+1}{14+20}$$

$$P(\text{"predictable"}|+) = \frac{0+1}{9+20}$$

$$P(\text{"no"}|+) = \frac{0+1}{9+20}$$

$$P(\text{"fun"}|+) = \frac{1+1}{9+20}$$

2. Scoring the test set

$$P(-) \cdot P(S|-) = \frac{3}{5} \times \frac{2 \times 2 \times 1}{24^3} = 6.1 \times 10^{-5}$$

$$P(-) P(S|+) = \frac{2}{5} \times \frac{1 \times 1 \times 2}{293} = 3.27 \times 10^{-5}$$

System should assign Negative class

4. 1. compute $P(\text{cats} | I, \text{like})$
 count trigram = (1 like cats) = 1
 count bigram = (1 like) = 2
 $P(\text{cats} | 1 \text{ like}) = \frac{1}{2} = 0.5$

2. $P(\text{dogs} | \text{you, like})$
 trigram count (you like dogs) = 0
 probability = 0
 bigram backoff
 count (like dogs) = 1
 total after "like" = 3
 $P(\text{dogs} | \text{like}) = \frac{1}{3}$

3. Some trigrams don't appear in small training data, without backoff their probability become 0 and model cannot predict new sentences.

Backoff lets the model use lower order probability so it can make reasonable predictions.

	cat	dog	rabbit
5. cat	5	10	5
dog	15	20	10
rabbit	0	15	10

$$\text{precision} = \frac{TP}{TP + FP}$$

10 precision & Recall

→ cat $TP = 5$

$$FP = 10 + 5 = 15$$

$$FN = 15 + 0 = 15$$

$$\text{Recall} = \frac{TP}{TP + FN}$$

$$\text{precision} = 5 / (5 + 15) = 0.25$$

$$\text{Recall} = 5 / (5 + 15) = 0.25$$

→ Dog

$$TP = 20$$

$$FP = 15 + 10 = 25$$

$$FN = 10 + 15 = 25$$

$$\text{precision} = 20 / (20 + 25) = 0.44$$

$$\text{Recall} = 20 / (20 + 25) = 0.44$$

→ Rabbit

$$TP = 10$$

$$FP = 0 + 15 = 15$$

$$FN = 5 + 10 = 15$$

$$\text{precision} = 10 / (10 + 15) = 0.4$$

$$\text{Recall} = 10 / (10 + 15) = 0.4$$

2. i Macro Average

$$\text{Precision} = (0.25 + 0.44 + 0.40) / 3 = 0.36$$

$$\text{Recall} = (0.25 + 0.44 + 0.40) / 3 = 0.36$$

ii. Micro Average

$$\text{Total TP} = 5 + 20 + 10 = 35$$

$$\text{FP} = 15 + 25 + 15 = 55$$

$$\text{FN} = 15 + 25 + 15 = 55$$

$$\text{Precision} = 35 / (35 + 55) = 0.39$$

$$\text{Recall} = 35 / (35 + 55) = 0.39$$

iii, Macro Average treats all classes equally and average their score

Micro Average: combines all predictions and gives more weight to large classes